

Practice Problems on Word Embedding and Transformer

CSE427: Machine Learning

Part A: Word Embedding Concepts

Practice Problem

Q1. Neural networks cannot directly process raw words. Explain how word embeddings solve this problem. In your answer, explain why assigning arbitrary numbers to words is not a good representation.

Practice Problem

Q2. Consider the words “great” and “awesome”. These two words are used in similar ways. Explain what should happen to their vector representations after training.

Practice Problem

Q3. The word “great” can be used positively in one sentence and sarcastically in another sentence. Explain why multiple embedding dimensions are useful for representing such words.

Practice Problem

Q4. Suppose a word embedding model is trained on the following sentences:

MovieA is great

MovieB is great

Explain why the embeddings of MovieA and MovieB may become similar after training.

Practice Problem

Q5. In a word embedding model, the final output prediction is not always the main thing we want. Explain what we actually want to obtain after training and why.

Part B: Vocabulary, One-Hot Encoding, and Embedding Matrix

Practice Problem

Q6. Given the vocabulary:

[MovieA, is, great, MovieB]

represent each word using one-hot encoding.

Practice Problem

Q7. Suppose a vocabulary has 4 words and each word should be represented using 2 embedding values.

- (a) What is the size of the embedding matrix?
- (b) How many trainable embedding parameters are there?
- (c) What does each row of the embedding matrix represent?

Practice Problem

Q8. Given the vocabulary order:

[MovieA, is, great, MovieB]

and the embedding matrix:

$$E = \begin{bmatrix} 0.38 & 0.42 \\ 0.10 & 0.90 \\ 1.85 & 0.98 \\ 0.36 & 0.40 \end{bmatrix}$$

find the embedding vector for each word.

Practice Problem

Q9. Using the same embedding matrix,

$$E = \begin{bmatrix} 0.38 & 0.42 \\ 0.10 & 0.90 \\ 1.85 & 0.98 \\ 0.36 & 0.40 \end{bmatrix}$$

show how the one-hot vector for “great” selects its embedding vector.

Practice Problem

Q10. If a model has vocabulary size $V = 10000$ and embedding dimension $d = 300$, calculate the embedding matrix size and total number of embedding parameters. Explain why this can become expensive.

Part C: Learning Word Embeddings

Practice Problem

Q11. From the sentence:

MovieA is great

construct possible training pairs for a next-word prediction task.

Practice Problem

Q12. Suppose a model gives the following probabilities for the next word:

$$P(\text{MovieA}) = 0.23, \quad P(\text{is}) = 0.27, \quad P(\text{great}) = 0.26, \quad P(\text{MovieB}) = 0.24$$

- (a) Which word will the model predict?
- (b) If the correct next word is “is”, is the model very confident?
- (c) What should training do to these probabilities?

Practice Problem

Q13. A model predicts the next word after “MovieA”. The correct next word is “is”. The output probabilities are:

$$\hat{y} = [0.23, 0.27, 0.26, 0.24]$$

where the vocabulary order is:

[MovieA, is, great, MovieB]

Compute the cross-entropy loss.

Practice Problem

Q14. Explain why full softmax becomes slow when the vocabulary is very large.

Practice Problem

Q15. In the training pair:

aardvark $\rightarrow$ *eats*

explain how negative sampling would train the model without updating every word in the vocabulary.

Part D: CBOW and Skip-Gram

Practice Problem

Q16. For the sentence:

MovieA is great

construct one CBOW training example.

Practice Problem

Q17. For the same sentence:

MovieA is great

construct one Skip-Gram training example.

Practice Problem

Q18. Compare CBOW and Skip-Gram. Your answer should include their input and target.

Part E: Positional Encoding

Practice Problem

Q19. A Transformer receives all tokens at once instead of reading them one by one. Explain why positional encoding is needed.

Practice Problem

Q20. For the sentence:

I will go to school

assign position indices to all words. Then explain how the model knows the position of the word “go”.

Practice Problem

Q21. The word “go” has embedding:

$$E_{go} = [e_1, e_2, e_3, e_4]$$

and positional encoding:

$$PE_{go} = [p_1, p_2, p_3, p_4]$$

Write the final input vector that will be given to the Transformer.

Practice Problem

Q22. For the word “go”, suppose:

$$pos = 2, \quad d_{model} = 4$$

Calculate the full positional encoding vector.

Use:

$$PE(pos, 2i) = \sin\left(\frac{pos}{10000^{2i/d_{model}}}\right)$$

$$PE(pos, 2i + 1) = \cos\left(\frac{pos}{10000^{2i/d_{model}}}\right)$$

and:

$$\sin(2) = 0.9093, \quad \cos(2) = -0.4161$$

$$\sin(0.02) = 0.019998, \quad \cos(0.02) = 0.9998$$

Practice Problem

Q23. Suppose:

$$E_{go} = [0.20, -0.50, 0.70, 1.00]$$

and the positional encoding of “go” is:

$$PE_{go} = [0.9093, -0.4161, 0.019998, 0.9998]$$

Find the final Transformer input vector for “go”.

Practice Problem

Q24. In the sentence:

Lets go

the token embeddings are:

$$Lets = [1.87, 0.09]$$

$$go = [-0.78, 0.27]$$

and the positional encodings are:

$$PE_{Lets} = [0.0, 1.0]$$

$$PE_{go} = [-0.9, 0.4]$$

Calculate the vectors that will enter the encoder.

Part F: Query, Key, Value

Practice Problem

Q25. Explain Query, Key, and Value using simple language. Your answer should make clear which one asks, which one is matched, and which one carries information.

Practice Problem

Q26. For a token representation X , explain how Q , K , and V are produced.

Practice Problem

Q27. A student says: “The same Query, Key, and Value matrices are used everywhere in the encoder and decoder.” Is the student correct? Explain.

Practice Problem

Q28. In encoder-decoder cross-attention, explain where Query comes from and where Key and Value come from.

Part G: Encoder Self-Attention

Practice Problem

Q29. The encoder receives:

$$X_L = [1.87, 1.09]$$

$$X_G = [-1.68, 0.67]$$

The encoder uses:

$$W_Q = \begin{bmatrix} 1.1 & 0.6 \\ -2.8 & 2.4 \end{bmatrix}$$

$$W_K = \begin{bmatrix} -1.7 & 0.5 \\ -1.5 & 0.9 \end{bmatrix}$$

$$W_V = \begin{bmatrix} 1.5 & -1.0 \\ -0.3 & -0.2 \end{bmatrix}$$

Compute the Query, Key, and Value vectors for both input tokens.

Practice Problem

Q30. Using the Query and Key vectors below:

$$Q_L = [-1.0, 3.7], \quad Q_G = [-3.7, 0.6]$$

$$K_L = [-4.7, 1.9], \quad K_G = [1.9, -0.2]$$

calculate the complete encoder attention score matrix.

Practice Problem

Q31. Given the encoder score matrix:

$$\begin{bmatrix} 11.7 & -2.6 \\ 18.5 & -7.2 \end{bmatrix}$$

explain what each row represents. Then approximate the attention weights using row-wise softmax.

Practice Problem

Q32. Given:

$$V_L = [2.5, -2.1]$$

$$V_G = [-2.7, 1.5]$$

and attention weights:

$$[1.0, 0.0]$$

for both tokens, compute the encoder attention outputs.

Practice Problem

Q33. The original encoder inputs are:

$$X_L = [1.87, 1.09]$$

$$X_G = [-1.68, 0.67]$$

and both attention outputs are:

$$[2.5, -2.1]$$

Compute the final encoder outputs after adding attention outputs to the original inputs.

Part H: Decoder Self-Attention

Practice Problem

Q34. The decoder starts with the token $\langle EOS \rangle$. Its embedding is:

$$[2.70, -1.34]$$

and its positional encoding is:

$$[0.0, 1.0]$$

Compute the decoder input vector.

Practice Problem

Q35. The decoder input vector is:

$$X_E = [2.70, -0.34]$$

The decoder self-attention matrices are:

$$W_Q = \begin{bmatrix} 0.4 & 0.4 \\ -0.3 & 0.1 \end{bmatrix}$$

$$W_K = \begin{bmatrix} 0.4 & -0.7 \\ -0.4 & 0.3 \end{bmatrix}$$

$$W_V = \begin{bmatrix} -1.1 & -0.7 \\ -0.4 & 1.3 \end{bmatrix}$$

Compute the decoder self-attention output for the start token.

Practice Problem

Q36. After decoder self-attention, the representation becomes:

$$[-0.1, -2.6]$$

Explain what this vector represents and why the decoder still needs cross-attention.

Part I: Encoder-Decoder Cross-Attention

Practice Problem

Q37. The decoder representation is:

$$E = [-0.1, -2.6]$$

The encoder outputs are:

$$L = [4.37, -1.01]$$

$$G = [0.82, -1.43]$$

The cross-attention matrices are:

$$W_Q = \begin{bmatrix} 1.5 & -0.3 \\ 0.3 & -1.0 \end{bmatrix}$$

$$W_K = \begin{bmatrix} -1.1 & 0.3 \\ -0.3 & -0.8 \end{bmatrix}$$

$$W_V = \begin{bmatrix} 1.1 & 0.6 \\ -1.2 & -0.5 \end{bmatrix}$$

Compute the cross-attention output.

Practice Problem

Q38. After cross-attention, the decoder vector becomes:

$$[5.9, 0.5]$$

Explain how this vector is used to generate the next output token.

Practice Problem

Q39. The output vocabulary is:

$$ir, \quad Vamos, \quad y, \quad \langle EOS \rangle$$

Suppose the dense layer and softmax give the highest probability to:

$$Vamos$$

What token is generated? What becomes the decoder input for the next step?

Part J: Second Decoder Step

Practice Problem

Q40. After generating “Vamos”, the decoder input contains:

$$\langle EOS \rangle, \quad V_{amos}$$

Given:

$$K_E = [1.2, -1.8], \quad V_E = [-2.8, -2.3]$$

$$Q_V = [-1.3, -0.1], \quad V_V = [0.0, 3.7]$$

and:

$$\text{softmax}(-1.1, 1.4) = [0.1, 0.9]$$

compute the self-attention output for “Vamos”.

Practice Problem

Q41. The original representation of “Vamos” is:

$$[-0.86, 2.37]$$

and its self-attention output is:

$$[-0.28, 3.10]$$

Compute the updated representation after self-attention.

Practice Problem

Q42. In the second decoder step, the cross-attention scores are:

$$[-10.3, -7.1]$$

and the attention weights are:

$$[0.0, 1.0]$$

The encoder value vectors are:

$$V_L = [6.0, 3.1]$$

$$V_G = [2.6, 1.2]$$

Compute the cross-attention output and the final updated representation of “Vamos” if the representation before cross-attention is:

$$[-1.2, 5.5]$$

Practice Problem

Q43. Suppose the decoder now predicts:

$\langle EOS \rangle$

Explain what happens to the generation process.

Part K: Full Problems

Practice Problem

Q44. Full word embedding problem

Given:

MovieA is great

MovieB is great

vocabulary order:

[MovieA, is, great, MovieB]

embedding matrix:

$$E = \begin{bmatrix} 0.38 & 0.42 \\ 0.10 & 0.90 \\ 1.85 & 0.98 \\ 0.36 & 0.40 \end{bmatrix}$$

and output probabilities for input MovieA:

[0.23, 0.27, 0.26, 0.24]

Answer the following without being told each formula separately:

- (a) Represent MovieA as a one-hot vector.
- (b) Find its embedding vector.
- (c) Identify the predicted next word.
- (d) Compute the loss if the correct next word is “is”.
- (e) Explain how this training process can make MovieA and MovieB similar.

Practice Problem

Q45. Full positional encoding problem

For the sentence:

I will go to school

the word “go” appears at position 2. Assume $d_{model} = 4$.

Compute the complete positional encoding vector for “go” and add it to the embedding:

$$E_{go} = [0.20, -0.50, 0.70, 1.00]$$

Use the sinusoidal positional encoding method from class notes.

Practice Problem

Q46. Full encoder self-attention problem

For input:

Lets go

the vectors entering the encoder are:

$$X_L = [1.87, 1.09]$$

$$X_G = [-1.68, 0.67]$$

The encoder matrices are:

$$W_Q = \begin{bmatrix} 1.1 & 0.6 \\ -2.8 & 2.4 \end{bmatrix}, \quad W_K = \begin{bmatrix} -1.7 & 0.5 \\ -1.5 & 0.9 \end{bmatrix}, \quad W_V = \begin{bmatrix} 1.5 & -1.0 \\ -0.3 & -0.2 \end{bmatrix}$$

Perform the complete encoder self-attention calculation and obtain the final encoder output vectors.

Practice Problem

Q47. Full decoder and cross-attention problem

The encoder outputs are:

$$L = [4.37, -1.01], \quad G = [0.82, -1.43]$$

The decoder starts with:

$$\langle EOS \rangle$$

whose decoder input vector is:

$$X_E = [2.70, -0.34]$$

Using the decoder self-attention and cross-attention calculations from class notes, compute the final decoder vector that is sent to the dense layer and softmax.

Practice Problem

Q48. Full Transformer explanation

Explain the complete Transformer encoder-decoder process for:

Input : Lets go

Output : Vamos

Your answer should include the mathematical role of:

embedding, positional encoding, Q, K, V , softmax attention, cross-attention, dense layer, $\langle E$